Anubhav Chakraborty, PhD

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Professional Summary

Molecular biologist and genetic engineer with a PhD in Molecular and Integrative Physiology and 7 years of research experience in functional genomics, epigenetic regulation, cellular engineering, gene delivery, and disease modeling. Expertise in CRISPR-based gene regulation and editing, primary mammalian cell models, viral and non-viral delivery systems, and development of experimental strategies to investigate mechanisms underlying cellular phenotypes. Proven ability to independently develop innovative approaches, troubleshoot complex biological problems, and translate mechanistic findings into therapeutic concepts. Strong record of scientific productivity including peer-reviewed publications, a pending U.S. patent, competitive fellowship funding, national presentations, multidisciplinary collaboration, and mentoring of junior researchers.

Education/Employment

University of Kansas Medical Center (KUMC) — **Kansas City, KS, USA**
Doctor of Philosophy (PhD) in Molecular and Integrative Physiology — *Aug 2019 – Aug 2026*
Cumulative GPA: 3.88/4.0

Vellore Institute of Technology — **Tamil Nadu, India**
Council of Scientific and Industrial Research (CSIR) - Govt. of India, — *Dec 2017 – Jun 2019*
Senior Research Fellow

SRM Institute of Science and Technology — **Tamil Nadu, India**
Master of Technology (MTech) in Genetic Engineering — *Jun 2015 – May 2017*
First Class (CGPA: 7.23/10)

SASTRA University — **Tamil Nadu, India**
Bachelor of Technology (BTech) in Bioengineering — *Jun 2011 – May 2015*
First Class (CGPA: 7.33/10)

Technical Skills

- **Functional Genomics & Molecular/Cell Biology:** CRISPR activation (CRISPRa) and interference (CRISPRi), CRISPR-Cas9/Cas12a gene editing, Chromatin accessibility assays, Advanced molecular cloning, ELISA, Immunoblotting, Immunocytochemistry, RT-qPCR, Fluorescent-reporter assays and screening.
- **Cellular engineering & Gene Delivery:** Mini-circle production, plasmid engineering and optimization, Protein trans-splicing systems, Recombinant AAV (rAAV) and lentivirus design and production, Virus-Like Particles (VLPs), Lipid Nanoparticles (LNPs), Nucleofection.
- ***In vitro* & *In vivo* Disease Models:** Primary human and mouse renal epithelial cell culture, Mouse ischemia-reperfusion injury model, Tissue microdissection, 3D cell culture, Cell line development, Primary cells isolation and engineering.
- **Computational Tools:** Benchling & SnapGene (i*n silico* vector design), LabArchive (Electronic Lab Notebook), GraphPad Prism, Microsoft Office Suite.

Research Experiences

The Jared Grantham Kidney Institute, University of Kansas Medical Center — **Kansas City, KS, USA**
Predoctoral candidate | Lab of Dr. Alan S. Yu — *Aug 2020 – Aug 2026*

- **Mechanistic gene regulation:** Developed and characterized a CRISPRa-based approach to activate endogenous *PKD1* transcription, investigating how targeted transcriptional modulation alters protein expression and rescues disease-associated cellular phenotypes; resulted in a U.S. provisional patent and peer-reviewed publications.

- **Epigenetic regulation:** Investigated chromatin accessibility and heterochromatin-associated limitations of transcriptional activation at endogenous *PKD1* gene promoter, developing experimental strategies that overcame epigenetic barriers to gene regulation; benchmarked various CRISPR-effectors for optimal epigenomic modulation.
- **Gene Delivery and Cell Engineering:** Designed, evaluated, and optimized diverse vector delivery systems including rAAVs, VLPs, Lentiviruses, Nucleofection and LNPs; for delivering complex genetic payloads across primary and immortalized cell lines to develop *in vitro* diseases models.
- **Plasmid Engineering:** Engineered an innovative plasmid miniaturization strategy that improved transgene delivery by ~3-fold in difficult-to-transfect primary cells, demonstrating independent problem-solving and development of novel experimental approaches.
- **Experimental strategy & disease modeling:** Designed and optimized *in vitro* cellular disease models, reporter-based guide RNA screening assays, split-intein protein trans-splicing mechanisms to investigate gene regulation, protein expression, cellular signaling, and disease-associated phenotypes.
- **Consulting & Mentorship:** Served as a subject-matter expert advising inter-departmental laboratories on bespoke viral vector design, plasmid engineering, and stable line derivation; directly trained and mentored 4 junior researchers (1 intern, 3 pre-med students) in experimental design and molecular workflows.

Vellore Institute of Technology (VIT) **Tamil Nadu, India**

Senior Research Fellow (CSIR) | Lab of Dr. Pragasam V *Dec 2017 – Jun 2019*

- **Epigenetic regulation & disease modeling:** Investigated DNA methylation dynamics associated with necroptotic cell death and established mouse ischemia-reperfusion injury models to investigate molecular mechanisms underlying tissue injury and disease progression.
- **Grant Proposals & mentoring:** Assisted faculty in technical drafting of competitive grant proposals; co-directed master's thesis execution for 4 graduate students.

Bachelor's and Master's Research Experience *Jul* 2014- *May* 2017

SRM Institute of Science and Technology *(Lab of Dr. J. Megala)* | Investigated genetic and transcriptional mechanisms underlying TNF-α promoter polymorphisms and gastric mucosal inflammation; genotyped 1,100 clinical subjects and performed phylogenetic/transcriptional analyses.

SASTRA University *(Lab of Dr. C. David Raj)* | Conducted phytochemical screening of various essential oils to isolate and evaluate the spasmolytic activity of phytochemical compounds using *ex vivo* superfusion assays on rat ileum

Patent, Selected Publications and Preprints (Full list PubMed Bibliography)

- **US patent pending: Chakraborty, A**., Wallace, D.P., and Yu, A.S.L. "CRISPRa Compositions, Methods of Use Thereof for Treating Polycystic Kidney Disease, and Methods of Making the Same." U.S. Provisional Patent Application No. 64/034,047, filed April 9, 2026.
- **Chakraborty, A**., Varghese, M.M, Wallace, D.P., Ward, C.J., Yu, A.S.L., 2026. CRISPR activation of endogenous PKD1 increases polycystin-1 levels and suppresses cellular features of ADPKD. bioRxiv. https://doi.org/10.64898/2026.05.23.727418
- **Chakraborty, A**. and Yu, A.S.L., 2026. Miniaturization of CRISPRa plasmids for efficient delivery into renal epithelial cells and Pkd1 transactivation. Molecular Biology Reports, 53 (707), https://doi.org/10.1007/s11033-026-11872-1
- Nyimanu, D., **Chakraborty, A**., Parnell, S., Wallace, D. and Yu, A., 2026. Apelin inhibits cyst growth and improves kidney function in mice with polycystic kidney disease. bioRxiv, pp.2026-03. https://doi.org/10.64898/2026.03.26.714294
- **Chakraborty, A**. and Yu, A.S.L., 2025. Prospects for gene therapy in polycystic kidney disease. Current Opinion in Nephrology and Hypertension, 34(1), pp.121-127. https://doi.org/10.1097/mnh.0000000000001030

Selected Fellowships, Honors & Awards (Full list on pages 3 & 4 –Digital Portfolio)

- **American Heart Association (AHA) Predoctoral Fellowship:** Awarded $71,828 (Jan 2024 – Dec 2025). https://doi.org/10.58275/AHA.24PRE1194472.pc.gr.190750
- **American Society of Nephrology (ASN) Kidney STARS & Physiology Society Travel Award:** ASN, Houston, TX (2025) & Philadelphia, PA (2023), KUMC (2023).
- **ASN Kidney TREKS Scholar:** Selected for MDBIL Research Program, Bar Harbor, ME (2024).
- **Biomedical Research Training Program Scholarship:** Awarded $13,000 – KUMC (2023).
  Best Poster Award: 4th Annual Internal Medicine Research Day, KUMC (2023).

Selected Scientific Presentations (Full list on pages 4 & 5 –Digital Portfolio)

Invited Oral Presentations:

- **Chakraborty, A**. and Yu, A.S.L. Miniaturized CRISPR activation plasmid improves transgene delivery, PKD1 expression and induces signaling correction in polycystic kidney cells. Jared J. Grantham Symposium, Kansas City, KS (2025).
- **Chakraborty, A**. and Yu, A.S.L. Epigenetic reprogramming of PKD1 transcription: A proof of concept. Jared J. Grantham Symposium, Kansas City, KS (2023).

Poster Presentations:

- **Chakraborty, A**., Wallace, D.P., Yu, A.S.L. Rationally engineered miniaturized plasmids improve transgene delivery and CRISPR-activation of PKD1 in primary kidney cells. ASN Kidney Week, Houston, TX (2025).
- **Chakraborty, A**., Ward, C.J., Yu, A.S.L. CRISPR activation of PKD1 in immortalized cell lines is limited by its heterochromatinized proximal promoter. ASN Kidney Week, Philadelphia, PA (2023).
- **Chakraborty, A**., Parnell, S.C., Zhang, Y., Daniel, E., Vrtek, K., Raman, A., Reif, G., Wallace, D.P. Early cyst formation leads to the development of an inflammatory microenvironment and tissue remodeling in polycystic kidney disease. ASN Kidney Week, Orlando, FL (2022).

Academic Leadership (Full list on page 5 –Digital Portfolio)

- **Planning Committee Member | Polycystic Kidney Disease-Renal Resource Consortium (PKD-RRC) Symposium (Apr 2023)**
  - Coordinated speaker scheduling, cross-institutional participation, and technical execution for a national scientific symposium.
- **Vice President | Cell Biology & Physiology Society, KUMC (Dec 2021 – Dec 2022)**
  - Led graduate-student programming include scientific seminars, professional-development workshops, and networking events.
- **Interprofessional Committee Member | Student Governing Council, KUMC (Aug 2022 – Aug 2024)**
  - Represented graduate researchers across health-profession schools and contributed to research-resource and grant discussions.
- **Organizing Committee Member | 3-Minute Thesis, Student Research Forum, KUMC (Jan 2022 – Dec 2024)**
  - Coordinated annual university-wide science-communication competitions, including judge recruitment, participant workshops, and event execution.